

RFQ - WSL-LANE-HARNESS-A Rev 4

RFQ — Custom Wire Harness Assembly (Lane Controller Field Harness)

Westside Lanes · Olympia, WA · Issued 2026-07-25 · Contact: Dylan DeYoung **Assemblies: WSL-LANE-HARNESS-A** Rev 4 · **WSL-PI-LINK-B** Rev 1 (*NEW — see §16*) **Quantity:** 1 first article → 2 pilot → balance to **34 total** (32 lanes + 2 spares). Option pricing requested at 48 and 64.

THIS RFQ NOW COVERS TWO ASSEMBLIES. Everything from §0 to §15 describes **WSL-LANE-HARNESS-A** (the machine field harness, 53 leads, one per LANE, qty 34). **§16 adds WSL-PI-LINK-B** — a small in-enclosure signal harness, 22 leads, one per ENCLOSURE, qty 18. Please quote them **separately**, and also tell us the NRE effect of taking both together (§16.6).

Rev 4 supersedes Rev 3 — quote against Rev 4 only. Neither Rev 2 nor Rev 3 was issued. **Rev 4 change: four new leads (W50–W53), two power pairs.** We are removing a legacy interface board from the machine and powering its sensors and a camera ourselves. **WARNING: The two pairs are electrically SEPARATE domains and must never be cross-connected** — see §6 and the wire-list notes. New colours **Violet/Grey** (isolated pair) and **Brown/Pink** (camera pair) are used for no other lead. Lead count 49 → 53.

Rev 3 change: the enclosure-to-machine run was measured at **6 ft (1829 mm)**. Rev 2 lead lengths were sized before that and were too short once the in-enclosure route is added. L1 2500→**3200**, L2 3000→**3700**, L3 4000→**4700**, L1-200 2300→**3000**. L0 (800) and the J14 jumper (120) are unchanged.

Rev 2 changes vs Rev 1 (all carried forward into Rev 3): (1) **All lead lengths increased** — Rev 1 budgeted the machine run but not the in-enclosure route (§3, §9). (2) **J14 ferrules corrected to bare/uninsulated** — an insulated 0.75 mm² collar fouls the adjacent pole at 3.5 mm pitch (§5). (3) **Terminal torque now specified** (§5). (4) **Strip length corrected 8 mm → 7 mm** (§5). (5) **Coding profiles added** to Tier 1 (§4). (6) **1840489 is customer-free-issued** (§4). (7) Two questions we previously asked you are now answered by us (§Questions).

0. Read this first — what kind of customer we are

We are a **small business (a bowling center), not an OEM**. We do not require PPAP, a formal FAI package, ISO quality-system flow-down, a supplier audit, or source inspection. **Your standard Certificate of Conformance is acceptable.** Please do not price a documentation program into this quote — if you see a line item that exists only to satisfy an aerospace or automotive customer, leave it out and tell us you did.

What we *do* care about, in priority order: **correct and unambiguous labeling**, 100% electrical test, consistency unit-to-unit, and on-time delivery.

1. What this assembly is

A field wiring harness for a replacement control board on bowling pinsetter machines. One harness per lane. It connects a controller PCB (mounted in a wall enclosure) to sensors, switches, and relay-coil circuits on the machine. The enclosure-to-machine run is **measured at 6 ft (1.83 m)**; leads are built longer than that to cover the in-enclosure route plus service slack.

Electrically trivial — nothing above 33 V, nothing above ~1 A. This is a **labeling- and termination-intensive** assembly, not a high-voltage or high-current one. Cost lives in the ~53 leads × 2 ends of printed markers, not in the wire.

Environment: indoor, unconditioned equipment area behind bowling pinsetters. Ambient to ~35 °C. Fine wood/pin dust and light lane-oil mist. Vibration from adjacent machinery. No flexing in service once installed — this is a static installation.

2. Scope boundary (important)

Quote the **controller-side harness only**, as specified here:

- **Controller end (one end of every lead): FULLY TERMINATED** — insulated ferrule, landed and torqued into its Phoenix plug, per \$4/\$5.
- **Machine end (the other end): PREPARED BUT UNTERMINATED** — cut to length, labeled, and left with **10 mm of insulation stripped, no ferrule** unless noted otherwise in the wire list. We terminate the machine end ourselves on site (the machine-side interface is a legacy 1982 connector still being characterized).

Do **not** quote any machine-side connector. If you have an opinion on whether the stripped ends should instead ship unstripped, say so.

3. Wire list

Attached as a separate CSV (`WSL-LANE-HARNESS-A_wirelist_rev4.csv`), one row per lead, with columns: `Wire ID` • `AWG` • `UL style` • `Color` • `End-A connector ref + position` • `End-A termination` • `End-B termination` • `**Length Class**` • `Finished length` • `Label text (identical both ends)` • `Twisted-pair partner` • `Notes` .

Rev 2 added a Length Class column; Rev 3 revises the class values. Every lead belongs to one of four classes (**L0 / L1 / L2 / L3**). If we revise lengths again, only the class values change — the per-lead assignments stay put. Please structure your quote so a class-length change is a price delta, not a re-quote.

Summary for scoping:

Group	Connector (controller end)	Leads	AWG	Length class	Length
Fast field inputs — cams	J3 — Phoenix MC 1,5/10-ST-3,5	5 (positions 1–5; all capped)	22	L3	4.7 m
Fast field inputs — DIELL	J3 (positions 7–10; 6 empty)	4	22	L2	3.7 m
Slow inputs A	J4 — MC 1,5/14- ST-3,5	13 (positions 1–11, 13, 14; 12 empty)	22	L1	3.2 m
Slow inputs B	J5 — MC 1,5/12- ST-3,5	4 (positions 1, 2, 3, 12; 4–11 empty)	22	L1	3.2 m (FOUL lead 3.0 m)
Lamp outputs	J13 — MC 1,5/ 6- ST-3,5	6	22	L3	4.7 m (see §9)
Safety loop	J14 — MC 1,5/ 4- ST-3,5	2 + 1 internal jumper	18	L1	3.2 m (jumper 120 mm)
Power in	none — ferruled loose leads	3	18	L0	0.8 m
Machine outputs	none — ferruled loose leads	12	18	L1	3.2 m
Sensor power (isolated) WARNING:	none — ferruled loose leads	2	18	L2	3.7 m
Camera power WARNING:	none — ferruled loose leads	2	18	L3	4.7 m
Total		~53 leads			

Length class definitions: **L0** = stays inside the enclosure · **L1** = enclosure to the machine connector · **L2** = enclosure to the photoelectric sensors · **L3** = enclosure to the overhead lamp unit / far mechanism.

4. BOM

Tier 1 — NO SUBSTITUTIONS (form/fit/function critical)

Item	Manufacturer	Part number	Qty/assy	Source
Plug, 10-pos, 3.5 mm	Phoenix Contact	1840447 (MC 1,5/10-ST-3,5)	1	you
Plug, 14-pos, 3.5 mm	Phoenix Contact	1840489 (MC 1,5/14-ST-3,5)	1	WARNING: CUSTOMER FREE-ISSUE
Plug, 12-pos, 3.5 mm	Phoenix Contact	1840463 (MC 1,5/12-ST-3,5)	1	you
Plug, 6-pos, 3.5 mm	Phoenix Contact	1840405 (MC 1,5/ 6-ST-3,5)	1	you
Plug, 4-pos, 3.5 mm	Phoenix Contact	1840382 (MC 1,5/ 4-ST-3,5)	1	you
Coding profile	Phoenix Contact	1734634 (CP-MSTB)	2	you

****WARNING:** 1840489 is FREE-ISSUED — do not source it.** It is discontinued and the authorized distribution channel is dry. We are making a lifetime buy and will ship it to you with the PO. Tell us your required buffer quantity. **Pre-approved fallback** if our stock runs short: **Phoenix FMC 1,5/14-ST-3,5** (push-in rather than screw; mates the identical MCV 1,5-G header, no other change). Use it only if we authorize in writing.

Coding profiles (new in Rev 2) — install 2 per assembly: J3 at pole 1 and J13 at pole 1. Profiles fit the **PLUG**, never a header. This is not cosmetic: the board has two pairs of *identical* connectors (J3/J15 both 10-pos, J13/J16 both 6-pos). An uncoded plug mates either one. A J13 lamp plug pushed into J16 lands LEDs across 5 V and wedges the I²C bus. Also apply the **WHITE identification band** per the marking scheme in the attached harness BOM. **Prove the coding operation on a sacrificial plug before coding production parts** — profile installation is not reversible.

Tier 2 — functional equivalents acceptable (tell us what you'd use)

Hook-up wire (UL1007 or UL1015 class, 300 V min for 22 AWG, 600 V for 18 AWG), insulated ferrules, printed heat-shrink marker stock, adhesive-lined heat-shrink end caps, lacing/ties, bags.

Wire colors are functionally meaningful — do not substitute colors. See §6.

5. Ferrule, strip, and torque specification

Strip length is 7 mm on every landed end. (Rev 1 said an 8 mm ferrule barrel — that was wrong; 7 mm is the Phoenix MC 1,5 figure.)

Leads	Lands in	Pitch	Ferrule
22 AWG → J3, J4, J5, J13	MC 1,5-ST-3,5 plugs	3.5 mm	0.34 mm² INSULATED , 7 mm
18 AWG → J14 only (JMP1, W33, W34)	MC 1,5/4-ST-3,5	3.5 mm	WARNING: BARE STRANDED, or 0.75–1.0 mm² UNINSULATED , 7 mm
18 AWG → loose leads for J2 / J6–J11	MKDS 1,5 blocks	5.08 mm	0.75 mm² INSULATED , 7 mm

****WARNING:** Why J14 is different — this is the one place Rev 1 was wrong.** The Phoenix MC 1,5 series is rated 0.14–1.5 mm² **bare or with an uninsulated ferrule, but only 0.5 mm² maximum with an insulated ferrule**. A 0.75 mm² insulated collar is roughly 4.0 mm across, against a 3.5 mm pole pitch — and all four J14 poles are populated, so adjacent collars physically interfere. **Do not fit insulated ferrules on any J14 lead**. The 0.75 mm² insulated ferrules on the loose J2/J6–J11 leads are correct and must stay: those blocks are 5.08 mm pitch, rated to 1.5 mm². Please do not “standardize” the three J14 leads to match them.

Terminal screw torque: 0.22–0.25 N·m (M2 screw). This is roughly **36 landed screws per assembly**. Do not use ~0.5 N·m — it is more than 2× the rating and will strip the screw or crack the plug body. Use a calibrated driver.

- Phoenix AI-series or Weidmüller H-series preferred; equivalents acceptable.
- Crimp to the ferrule manufacturer’s die spec; **every ferruled joint must pass a pull test**.

6. Wire colors (functional, not cosmetic)

Color	Meaning
White	landed sense signal
Orange	DO-NOT-LAND lead — capped at the machine end, deliberately unconnected until a future commissioning step. Orange must not appear on any other lead.
Green	field ground / chassis bond
Black	ground / return
Blue, Yellow (22 AWG)	photoelectric sensor signals L and R
Red (18 AWG)	relay normally-open output, and +5 V power
Black (18 AWG)	relay common, and power return
Yellow (18 AWG)	safety-loop circuit
Violet	isolated +12 V — sensor supply. WARNING: separate electrical domain
Grey	isolated 0 V — sensor return. WARNING: separate electrical domain
Brown	camera +12 V (non-isolated)
Pink	camera 0 V (non-isolated)

WARNING: The Violet/Grey pair and the Brown/Pink pair are two SEPARATE electrical domains and must never be joined, spliced, or share a ferrule. They are deliberately given four unique colours, used on no other lead, so the distinction survives installation. Please keep them in **separate bundles** and do not lace them together.

7. Labeling — the cost driver, specified precisely

- **Printed heat-shrink sleeve markers** (Brady PermaSleeve, Phoenix THERMOMARK, or equivalent), white sleeve, black print, permanent. Not wrap-around adhesive labels; not hand-marked.
- **BOTH ends of every lead**, ~53 leads = **~106 markers per assembly**.
- Exact text is in the wire-list CSV, column `Label text`. Text is identical at both ends of a given lead. **Labels are deliberately short — longest string is 22 characters** (`STOP/CIS - DO NOT LAND`). Signal names only; no machine-specific wiring references appear on the wire, because those vary per lane and live on a per-lane card instead. If any string is still too long for your marker stock, tell us your maximum — do not truncate on your own initiative.
- Position: within 40 mm of each termination, oriented so text reads with the lead running left-to-right.
- Additionally: each of the five Phoenix plug bodies carries its own printed label (text in the CSV, `Plug body label` sheet).

8. Special construction features

1. **Twisted pairs (2 per assembly)**. Blue+Black and Yellow+Black on J3, ~1 twist per 30–40 mm, starting 50 mm from the plug and stopping **100 mm short of the machine ends** so those ends remain independently routable. Pair partners are named in the wire list.
2. **Capped leads (11 per assembly)**. All orange leads plus the two 18 AWG yellow safety leads get their **machine end sealed with adhesive-lined heat shrink** — no ferrule, no exposed conductor. They still carry their printed label. These are deliberately dead-ended.
3. **Empty connector positions**. J3-6, J4-12, and J5 positions 4–11 are **intentionally empty**. Do not populate them. The wire list is authoritative; if a position isn't listed, it stays empty.
4. **J14 internal jumper**. A ~120 mm 18 AWG yellow U-jumper is fitted **inside the J14 plug between positions 1 and 2**, ferruled both ends, with a printed flag label on the loop (text in the CSV). This is a deliberate engineered link, not an error. Positions 3 and 4 carry the two capped leads. **A finished J14 must read SHORT across 1–2 and OPEN across 3–4**.
5. **Loose ferruled leads**. The 3 power leads and 12 machine-output leads have **no connector at the controller end** — they ship ferruled and labeled, loose, to be landed in screw terminals at install. Bundle and tie them per group.

9. Lengths — build long, we trim on site

Rev 3 lengths are measured-plus-margin. The enclosure-to-machine run is **6 ft (1829 mm)**, measured. Each L1 lead must also cross the in-enclosure route (~750–950 mm on the far board) plus ~300 mm service slack and drip loop, which is how 3200 mm is derived. Per-lane siting is not yet surveyed at every lane pair, so all four classes are sized so that **no lead can come up short**. We will trim on site. Build to the wire list; do not optimize lengths down.

****WARNING:** Marker placement requirement (new in Rev 2, and it affects your process).** Because we trim the machine end in the field, a marker printed at the very end of the lead gets cut off. On every **L1 / L2 / L3** lead, either: **(a)** position the machine-end marker **150 mm back from the cut end**, or **(b)** ship the machine-end markers loose, unshrunk, in a labeled bag per assembly. **State which you will do**. The controller-end marker is unaffected. (Correct labeling is priority #1 in §0 — a trimmed-off marker defeats the whole spec.)

Please quote: **(a)** at the Rev 3 lengths as written, and **(b)** the per-assembly cost delta per **±0.5 m on class L1** (30 leads), so we can price the classes down once the site survey is complete without re-quoting.

10. Test and workmanship

- **Workmanship: IPC/WHMA-A-620, Rev E, Class 2.** Class 2 is correct for this application — please do not build or price to Class 3.
- **100% electrical test on every assembly:** point-to-point continuity on all nets, **plus isolation/shorts testing between all nets.** Isolation matters more than usual here — an adjacent-position short inside a Phoenix plug is our most likely failure mode.
- **Explicit exception:** the J14 1–2 jumper is an intentional short and must not be flagged.
- **No hipot required** (max circuit voltage 33 V).
- Ship a test record or pass/fail log per serialized assembly.

11. Identification and traceability

- Each assembly carries a printed label with **assembly PN + revision + serial number** on the main bundle near the J4 plug.
- Serial-level traceability requested (simple sequential numbering is fine — we need to tie a harness to a lane, not to a wire lot).

12. Quantity, ramp, and spares

Stage	Qty	Note
First article	1	Ship for our approval before proceeding. We will respond within 5 business days.
Pilot	2	After FA approval
Balance	31	32 lanes + 2 spares total
Total program	34	Please also quote 48 and 64 so we can see the price break

Additionally quote as separate line items: **10% spare loose leads** (assorted, pre-labeled) and **3 spare sets of the five Phoenix plugs**, unpopulated.

13. Packaging and kitting

Individually bagged and labeled per assembly, with the serial number visible on the outside of the bag. This is worth paying for on a 32-lane install — it is what prevents a wrong-harness-on-wrong-lane afternoon. Coil to ~250 mm diameter, tie in at least 3 places, one carton per 6–8 assemblies with a packing list.

14. Commercial

- **Target:** first article within 3 weeks of PO; balance within 8 weeks of FA approval.
- Ship to Olympia, WA 98502. Freight prepaid and added is fine.
- Quote NRE separately from unit price, and tell us what the NRE buys (fixture, program, setup) and whether it recurs on a reorder.

- State your acceptable over/under shipment tolerance.
- **Drawing revision control stays with us.** We will issue revised wire lists as `Rev 2` , `Rev 3` , etc.; please quote against the revision stated on the PO.

15. Attachments provided with this RFQ

1. `WSL-PI-LINK-B_wirelist_rev1.csv` — the 22-lead wire list for the SECOND assembly (§16)
2. `WSL-LANE-HARNESS-A_wirelist_rev4.csv` — the 53-lead from-to wire list with label text and length classes. **This is the controlling document. Where it and this RFQ disagree, the CSV wins.**
3. `phase8_revD_harness_bom.csv` — plug + coding-profile part identities and the band-marking scheme
4. Phoenix Contact datasheets for the five Tier-1 plugs and the CP-MSTB coding profile

Not attached, and deliberately so: Rev 1 referenced a dimensioned 2D layout PDF and a golden-sample photo set. **Neither exists** — the prototype unit has not been built. Both references are withdrawn rather than left dangling. If a flat layout drawing would materially reduce your quote, say so and we will produce one.

Plug-body labels: the J14 plug body carries `J14 SAFE LOOP - SHORT 1-2` (text is in the wire list). The other four plug bodies need identification text we have **not** yet specified — quote plug-body labeling as a **separate line item** and we will supply the text on PO.

16. SECOND ASSEMBLY — WSL-PI-LINK-B (Rev 1)

A small, entirely **in-enclosure** signal harness. It links each controller board’s IDC20 breakout to the Raspberry Pi’s GPIO breakout, inside the same box. It never leaves the enclosure, sees no machine voltage, and carries nothing above 3.3 V.

Wire list attached: `WSL-PI-LINK-B_wirelist_rev1.csv` .

16.1 What it is

Leads	22 (11 per controller board × 2 boards)
Wire	22 AWG UL1007 300 V
Both ends	Screw-terminal blocks — ferruled both ends , 0.34 mm ² insulated, 7 mm strip
Finished length	600 mm, ALL 22 leads — uniform (see §16.3)
Colours	Black = ground. White = everything else. Deliberately minimal — see §16.4
Quantity	1 per enclosure · 16 enclosures + 2 spares = 18

Note the quantity differs from assembly A: **A is per LANE (34), B is per ENCLOSURE (18).**

16.2 WARNING: Scope boundary — one end is deliberately unassigned

- **End A (the C7 breakout end): FULLY DEFINED.** Every lead’s terminal number is fixed in the wire list and will not change.

- **End B (the Pi breakout end): FERRULED AND LABELLED, BUT NOT ASSIGNED TO A TERMINAL.**

We are still finalising which Pi GPIO carries which signal. Rather than hold this quote, every lead is **identified by its signal name on a printed marker at both ends** and we land End B ourselves on site from a wire-map card. This is the same approach we use for the machine end of assembly A, and it is why §16.3 specifies one uniform length.

Nothing about this is unusual for you — both ends are still ferruled to the same spec, and the assembly is fully testable end-to-end because the wire list defines every lead’s identity.

16.3 Why every lead is the same length

Because End B is unassigned, **any lead must be able to reach any terminal**. A mixed-length build would create leads that physically cannot serve their eventual destination. 600 mm covers the worst routed path in our enclosure with service slack; we trim on site.

Please do not optimise these to differing lengths. Uniformity is the requirement.

16.4 Colours — intentionally minimal, and why

Assembly A’s colours are functionally meaningful and tightly specified (§6). Assembly B deliberately uses **only Black and White** so that **no colour on B can ever be mistaken for a coded colour on A** if both assemblies are ever open in the same enclosure at the same time — which they will be.

Identification on B is by **printed marker only**. 22 leads × 2 ends = **44 markers**.

16.5 The one construction requirement that is not optional

Four twisted pairs, two per controller board. Each I²C signal must be twisted with its own dedicated ground return, as paired in the wire list:

Pair	Twist
P01 ↔ P02	board A ground ↔ board A SDA
P03 ↔ P04	board A SCL ↔ board A ground
P12 ↔ P13	board B ground ↔ board B SDA
P14 ↔ P15	board B SCL ↔ board B ground

This is electrical, not cosmetic. These are I²C buses running past switched relay coils, and one of the two buses is bit-banged in software and therefore more timing-fragile than a hardware bus. Roughly 1 twist per 25–35 mm is what we are after. If your standard practice differs, tell us what you do rather than dropping the twist.

16.6 Quantities and what we would like to know

Stage	Qty
First article	1
Pilot	1
Balance	16
Total	18

Please tell us: 1. Unit price at 18, and at 1 (we will buy a first article either way). 2. **Does taking BOTH assemblies together reduce total NRE**, versus quoting A alone? If some setup carries across, we would rather know now than add B later. 3. Whether the uniform-length, unassigned-End-B approach causes you any difficulty. If a defined End B would be materially cheaper, say so — we may be able to freeze it sooner.

Questions we specifically want answered in your quote

1. Your maximum legible characters per heat-shrink marker (\$7).
2. The \$9 length delta (per ± 0.5 m on class L1), and **which marker-placement option you will use** — set back 150 mm, or shipped loose and un-shrunk.
3. Your required **buffer quantity** for the free-issued 1840489 (\$4).
4. Confirmation that you can install **CP-MSTB coding profiles** (\$4), and whether you would rather we ship the plugs pre-coded.
5. Anything in this package that is over-specified and costing us money for no benefit. We would rather hear that than pay for it.

Two Rev-1 questions are now answered by us — no response needed: Ferrules (old Q1): resolved in §5. J14 takes bare or uninsulated; everything else is as tabulated. Do not substitute. **Machine-end preparation (old Q4):** build them **stripped 10 mm as specified**. The machine end is crimped into a contact on our side, so stripped is correct.